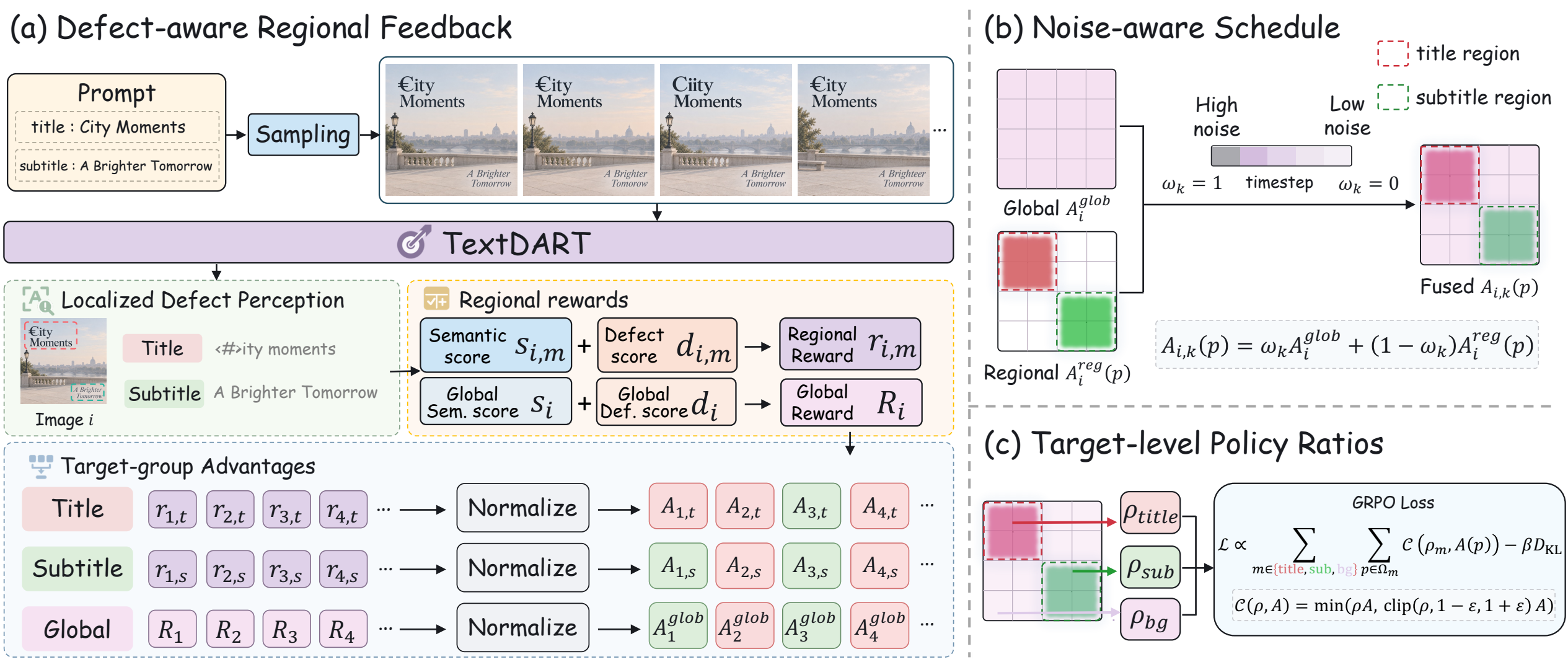

(a) Defect-aware Regional Feedback
Prompt
title : City Moments
subtitle : A Brighter Tomorrow
Sampling
TextDART
Localized Defect Perception
Title
<#>ity moments
Subtitle
A Brighter Tomorrow
Image i
Regional rewards
Semantic score $s_{i,m}$
Defect score $d_{i,m}$
Regional Reward $r_{i,m}$
Global Sem. score s_i
Global Def. score d_i
Global Reward R_i
Target-group Advantages
Title
Subtitle
Global
Normalize
Normalize
Normalize
(b) Noise-aware Schedule
title region
subtitle region
High noise
Low noise
$\omega_k = 1$
timestep
$\omega_k = 0$
Global A_i^{glob}
Regional $A_i^{reg}(p)$
Fused $A_{i,k}(p)$
$A_{i,k}(p) = \omega_k A_i^{glob} + (1-\omega_k) A_i^{reg}(p)$
(c) Target-level Policy Ratios
ρ_{title}
ρ_{sub}
ρ_{bg}
GRPO Loss
$\mathcal{L} \propto \sum_{m \in \{title, sub, bg\}} \sum_{p \in \Omega_m} \mathcal{C}(\rho_m, A(p)) - \beta D_{KL}$
$\mathcal{C}(\rho, A) = \min(\rho A, \text{clip}(\rho, 1-\varepsilon, 1+\varepsilon) A)$